

DS8003 Final Project Proposal

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Group Members

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1. Problem Definition

Cancer research is about going through different types of cancers and how it relates to patient's backgrounds along with the cancer stages, preventive steps and possible treatments. Effective treatments are hard for people to identify because cancers vary widely from lung, breast, skin, bladder, etc. It becomes difficult for researchers, healthcare and professionals to classify and identify clear patterns from it. Finding prevention strategies becomes difficult for healthcare professionals and also for the researchers. Sometimes it may hide important points/data with directly affect on therapies.

Due to this situation here, it becomes difficult to:

- Determine which cancers cause the most deaths and cases.
- Cancers spread across different tissues.

2. Suggested Solution

In our project, we will build a distributed Big Data pipeline for analyzing a Cancer Knowledge Graph.

The steps are as followed:

- Data Ingestion and Storage: Apache NiFi for general purpose ingestion of data, transformation, and routing through pipeline along with scalability. HDFS (Hadoop Distributed File System) is used for distribution and also for fault tolerance.

- Data Cleaning and Preprocessing: Hive for SQL querying and also for resolving missing values and to get data in structured and clean manner.
- Exploratory Data Analysis (EDA): Hive and Pig for pattern analyzing for treatments which is associated with specific cancer types and its effect on each stage.
- Graph Analysis and Feature Engineering: Pig for data flow processing and also for extracting relationships, deriving new features like success rate of treatment, and for mapping.
- Insight Generation: Through different process of aggregation, filtering, and basic modeling we can get the pipeline which give us at least five key insights.
- Evaluation and Visualization: Using statistics analysis and visual representations for understanding the insights properly and it can be used by professionals and researchers.

3. Dataset Description

Key Attributes to Use in Analysis:

- title – Gives cancer type/cancer center i.e. Breast Cancer, Lung Cancer.
- type – Classifies entry as cancer type/center.
- tissue – Affected tissue i.e. Lung, Skin, Breast.
- mainType – Classification of cancer i.e. Breast Cancer.
- oncoTreeld / parentOncoTreeld – Hierarchical identifiers used for cancer taxonomy.
- country, city – Geographic information for cancer centers.
- uscsCasesPerYear / uscsDeathsPerYear – Cancer incidence data.

The cancer dataset(<https://www.kaggle.com/datasets/sujaykapadnis/cancer-database>) is in .csv format and of size 470.16 kb which has 1453 entries and 96 attributes which include the cancer type and other key features that are used for understanding and analysis of the data.

References

1. Sujay Kapadnis. Cancer Database.

Available at: <https://www.kaggle.com/datasets/sujaykapadnis/cancer-database>